SOUGANDHI MANONMANI PENDYALA

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Summary

Software Development Engineer at Amazon experienced in building scalable full-stack applications, distributed systems, and cloud-native services using Java, Spring Boot, React, TypeScript, and AWS. Skilled in event-driven architectures, REST APIs, automated testing, and operational monitoring, with a strong foundation in software engineering, distributed systems, data structures, and algorithms.

Skills

- **Languages:** Java, TypeScript, JavaScript, Python, SQL, HTML, CSS
- **Full Stack:** React, Redux, TypeScript, Spring Boot, REST APIs, Microservices, Distributed Systems
- **Cloud & AWS:** AWS CDK, SQS, DynamoDB, CloudWatch, IAM, AppConfig
- **AI & LLMs:** Claude API, OpenAI API, Tool Calling, AI Agents, Prompt Engineering, TensorFlow.js
- **Devops & Tools:** Docker, Kubernetes, CI/CD, Git, GitHub, GitLab, Jira, Claude, Kiro
- **Testing:** JUnit, Mockito, Vitest, Integration Testing, Functional Testing
- **Soft Skills:** Problem Solving, Communication, Cross-Functional Collaboration, Agile Methodologies
- **Core CS:** Data Structures, Algorithms, Object-Oriented Programming

Professional Experience

Software Development Engineer | *Amazon, Detroit, MI, USA* Sep '25 – Oct '26

- Shipped a self-service BIL migration workflow using Spring Boot REST APIs and React with merchant access validation, replacing an ops-mediated process; achieved 100% unit coverage with Vitest.
- Built the Expansion Preferences page end-to-end in React with Zustand, enabling self-service configuration of multiple offer expansion preferences and reducing turnaround time by 25%.
- Re-architected offer expansion from synchronous request-blocking to an event-driven SQS FIFO pipeline, adding ordered consumers, filter-chain routing, DLQ recovery, & tiered CloudWatch alarms.
- Built policy-driven traffic shaping on AWS AppConfig, giving on-call a config-push lever to change rate limits across regions in minutes, work that previously required a full service deployment
- Built a React cooldown mechanism with localStorage persistence and polling hooks, preserving workflow state across refreshes/session breaks and reducing duplicate submissions by 40%.
- Carried on-call for the service, resolving incidents, driving RCAs, and adding diagnostic logging and alarms that reduced time-to-diagnose by 30% for recurring failures.
- Cut Java microservice cold-start time through improvements across 3 layers: JVM tuning, startup automation, & deployment pipelines, improving service readiness & reducing initialization overhead.
- Used Claude and Kiro across 4 development workflows: implementation, test generation, debugging, and iteration to compress feature turnaround while maintaining production-quality code.

Education

Illinois Institute of Technology, Chicago, USA **CGPA: 4.0/4.0**
Master of Information Technology and Management *Aug '23 – May '25*

Kamala Institute of Technology and Science, Telangana **CGPA: 3.0/4.0**
Bachelor of Technology in Electronics and Communication Engineering

Projects

Improving Software Quality Framework Using AI Platforms: Published a peer-reviewed book chapter on AI-driven software QA: defect detection, risk prediction, workflow automation, Agile/DevOps.

Mail Scheduler Agent | **Python, FastAPI, Supabase, Claude, Gmail API**: Built a Claude agent that schedules emails from plain English; Supabase Edge Functions + pg_cron send via Gmail with retries.

Platepix3D | **Next.js, TypeScript, TensorFlow.js, Three.js, Tailwind**: Built a prototype turning dish photos into 3D menu models via MobileNet and react-three-fiber; added inventory forecasting.

Multi Language AI Chatbot | **Python, Streamlit**: Engineered a 5-language chatbot using OpenAI and Google Translate APIs, with secure authentication, and persistent chat history for user interactions.

Pawcare | **Java, Android Studio**: Developed a pet care app for grooming, vet appointments, and pet-sitting with scheduling and reminders; integrated Google Maps, Firebase, and Calendar APIs.